

1. (a) No, S_1 is not closed under scalar multiplication because for $c = -1$ and $v = (1, 0)^T \in S_1$, then $cv = (-1, 0)^T \notin S_1$.
- (b) No, S_2 is not closed under addition because for $v_1 = (1, 0)^T \in S_2$ and $v_2 = (0, 1)^T \in S_2$, then $v_1 + v_2 = (1, 1)^T \notin S_2$.

2. (a)
 - If U_c is a subspace of $\mathbb{R}^3$, then $(0, 0, 0)^T \in U_c$ and hence $c = 0^2 + 0^2 + 0^2 = 0$.
 - If $c = 0$, then $U_c = U_0 = \{(x, y, z)^T \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 0\} = \{(0, 0, 0)^T\}$ is the zero subspace of $\mathbb{R}^3$.

Thus, U_c is a subspace of $\mathbb{R}^3$ if and only if $c = 0$.

- (b)
 - If W_d is a subspace of $\mathbb{R}^2$, then $(0, 0)^T \in W_d$. So there exists $s_0 \in \mathbb{R}$ such that $2s_0 - 1 = 0$ and $-s_0 + d = 0$. Thus, $d = \frac{1}{2}$.
 - If $d = \frac{1}{2}$, then

$$W_d = W_{\frac{1}{2}} = \{(2s - 1, -s + \frac{1}{2})^T : s \in \mathbb{R}\} = \{t(-2, 1)^T : t \in \mathbb{R}\} = \text{span}(\{(-2, 1)^T\})$$

is a subspace of $\mathbb{R}^2$.

Thus, W_d is a subspace of $\mathbb{R}^2$ if and only if $d = \frac{1}{2}$.

3. (a) For any $u \in \mathbb{R}^m$,

$$u \in \text{col}(AB) \implies u = (AB)x \text{ for some } x \in \mathbb{R}^n \xrightarrow{y=Bx} u = Ay \text{ for some } y \in \mathbb{R}^n$$

$$\implies u \in \text{col}(A).$$

So $\text{col}(AB) \subset \text{col}(A)$.

- (b) (c): It follows from (a).
- ($\supset$):

$$\text{col}(A) = \text{col}(ABB^{-1}) \stackrel{(a)}{\subset} \text{col}(AB).$$

So $\text{col}(AB) = \text{col}(A)$.

- (c) Take $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, then $A_2 = EA_1$, where $E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and

$$\text{col}(A_1) = \text{span}(\{(1, 0)^T\}) \neq \text{span}(\{(0, 1)^T\}) = \text{col}(A_2).$$

4. (a) Disproof: Consider $k = n = 2$, $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_2 = v_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Then $v_3 \in \text{span}(\{v_1, v_2\})$ since $v_3 = 0 \cdot v_1 + v_2$. But v_1 does not belong to $\text{span}(\{v_2, v_3\})$ since all linear combinations of v_2, v_3 have the first entry 0.

(b) Proof: By the assumption,

$$\begin{aligned}v_{k+1} &= a_1v_1 + a_2v_2 + \cdots + a_kv_k, \text{ for some } a_1, \cdots, a_k \in \mathbb{R} \text{ and} \\v_{k+2} &= b_1v_1 + b_2v_2 + \cdots + b_{k+1}v_{k+1}, \text{ for some } b_1, \cdots, b_{k+1} \in \mathbb{R}.\end{aligned}$$

Then we have

$$\begin{aligned}v_{k+2} &= b_1v_1 + b_2v_2 + \cdots + b_kv_k + b_{k+1}(a_1v_1 + a_2v_2 + \cdots + a_kv_k) \\&= (b_1 + b_{k+1}a_1)v_1 + (b_2 + b_{k+1}a_2)v_2 + \cdots + (b_k + b_{k+1}a_k)v_k.\end{aligned}$$

This shows $v_{k+2} \in \text{span}(\{v_1, v_2, \dots, v_k\})$.

5. Consider the equation

$$au + bv = 0 \in \mathbb{R}^n$$

for scalars a and b . This can be rewritten as

$$\frac{a+b}{2}(u+v) + \frac{a-b}{2}(u-v) = 0.$$

Since $u+v$ and $u-v$ are linearly independent, we have

$$\frac{a+b}{2} = \frac{a-b}{2} = 0.$$

Hence $a = b = 0$. This implies u and v are linearly independent.